sw[15:0]

btn[4:0] clk

MPG1 MPG

btn

clk enable

RD2_id_ex_reg[31:0]

RTL_REG C

CED Q

ALURes_ex_mem_reg[31:0]

RTL_REG C

CED Q

RD2_ex_mem_reg[31:0]

RTL_REG C

CED Q

mem1

mem

MEMWrite

clken

ALUResIn[31:0] ALUResOut[31:0] Address[31:0] WriteData[31:0] ReadData[31:0]

WA_ex_mem_reg[4:0]

RTL_REG C

CED Q

ALURes_mem_wb_reg[31:0]

RTL_REG C

CED Q

MemData_mem_wb_reg[31:0]

RTL_REG C

CED Q

WA_mem_wb_reg[4:0]

RTL_REG C

CED Q

rez_i RTL_MUX S

S=1'b1 I0[31:0] S=default I1[31:0] O[31:0]

id1

id

ExtOp RegWrite

clken

ExtImm[31:0]

Instr[25:0] RD1[31:0] RD2[31:0] WA[4:0] WD[31:0] func[5:0] rd[4:0] rt[4:0] sa[4:0]

pcp4_if_id_reg[31:0]

RTL_REG C

CED Q

ExtImm_id_ex_reg[31:0]

RTL_REG C

CED Q

RD1_id_ex_reg[31:0]

RTL_REG C

CED Q

func_id_ex_reg[5:0]

RTL_REG C

CED Q

rd_id_ex_reg[4:0]

RTL_REG C

CED Q

rt_id_ex_reg[4:0]

RTL_REG C

CED Q

sa_id_ex_reg[4:0]

RTL_REG C

CED Q

pcp4_id_ex_reg[31:0]

RTL_REG C

CED Q

uc1

uc

ALUSrc Br_getz Br_gtz Branch ExtOp
Jump MemToReg MemWrite RegDst RegWrite

ALUOp[1:0]

Instr[5:0]

ue1

ex

AluSRC GETZ GTZ RegDst Zero

ALURes[31:0]

AluOp[1:0]

BranchAddress[31:0] Ext_Imm[31:0] Pc4[31:0] RD1[31:0] RD2[31:0]
func[5:0] rd[4:0] rt[4:0]

rwa[4:0]
sa[4:0]

Bgetz_id_ex_reg

RTL_REG C

CED Q

Bgtz_id_ex_reg

RTL_REG C

CED Q

GETZ_ex_mem_reg

RTL_REG C

CED Q

GTZ_ex_mem_reg

RTL_REG C

CED Q

Bgetz_ex_mem_reg

RTL_REG C

CED Q

Bgtz_ex_mem_reg

RTL_REG C

CED Q

Zero_ex_mem_reg

RTL_REG C

CED Q

PCSrc1_i__0 RTL_AND

I0

I1 O

PCSrc1_i RTL_AND

I0

I1 O

PCSrc0_i__0 RTL_AND

I0

I1 O

PCSrc0_i RTL_OR

I0

I1 O

PCSrc_i RTL_OR

I0

I1 O

BrAddr_ex_mem_reg[31:0]

RTL_REG C

CED Q

ifetch1

ifetch

clken
jump
pcsrc
rst

branchAddress[31:0]
instruction[31:0]

jumpAddress[31:0] pc4[31:0]

instruction_if_id_reg[31:0]

RTL_REG C

CED Q

Mem2R_id_ex_reg

RTL_REG C

CED Q

RegWr_id_ex_reg

RTL_REG C

CED Q

outpt_i

RTL_MUX

S=3'b000 I0[31:0] S=3'b001 I1[31:0] S=3'b010 I2[31:0] S=3'b011 I3[31:0] S=3'b100 I4[31:0] S=3'b101 I5[31:0] S=3'b110 I6[31:0] S=default I7[31:0]

O[31:0]

S[2:0]

Mem2R_ex_mem_reg

RTL_REG C

CED Q

RegWr_ex_mem_reg

RTL_REG C

CED Q

Branch_id_ex_reg

RTL_REG C

CED Q

MemWr_id_ex_reg

RTL_REG C

CED Q

SSD1
SSD

clk an[7:0] digits[31:0] cat[6:0]

Mem2R_mem_wb_reg

RTL_REG C

CED Q

RegWr_mem_wb_reg

RTL_REG C

CED Q

ALUOp_id_ex_reg[1:0]

RTL_REG C

CED Q

Branch_ex_mem_reg

RTL_REG C

CED Q

ALUSrc_id_ex_reg

RTL_REG C

CED Q

MemWr_ex_mem_reg

RTL_REG C

CED Q

RegDst_id_ex_reg

RTL_REG C

CED Q

an[7:0] cat[6:0]

led[15:0]